

Variation in whole brain connectomes with age: magnetic resonance histology in 13 strains of mice

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PSTR 380.15

INTRODUCTION

Diffusion MRI (dMRI) has become a valuable tool in quantitative assessment of age-related changes in brain morphology and cytoarchitecture. We have extended dMRI to mouse models using postmortem scans at spatial resolution up to 2.4 million times that of clinical dMRI [1,2]. Using these methods, we have demonstrated significant strain related variation in both the diffusion scalar metrics and brain wide connectivity [3], but those measurements covered only four strains, at one time point. The work described here extends that investigation to age related changes across thirteen different strains providing insight into the utility of dMRI metrics in following age-related changes across a population of animals with genetic variation comparable to that encountered in the clinical setting.

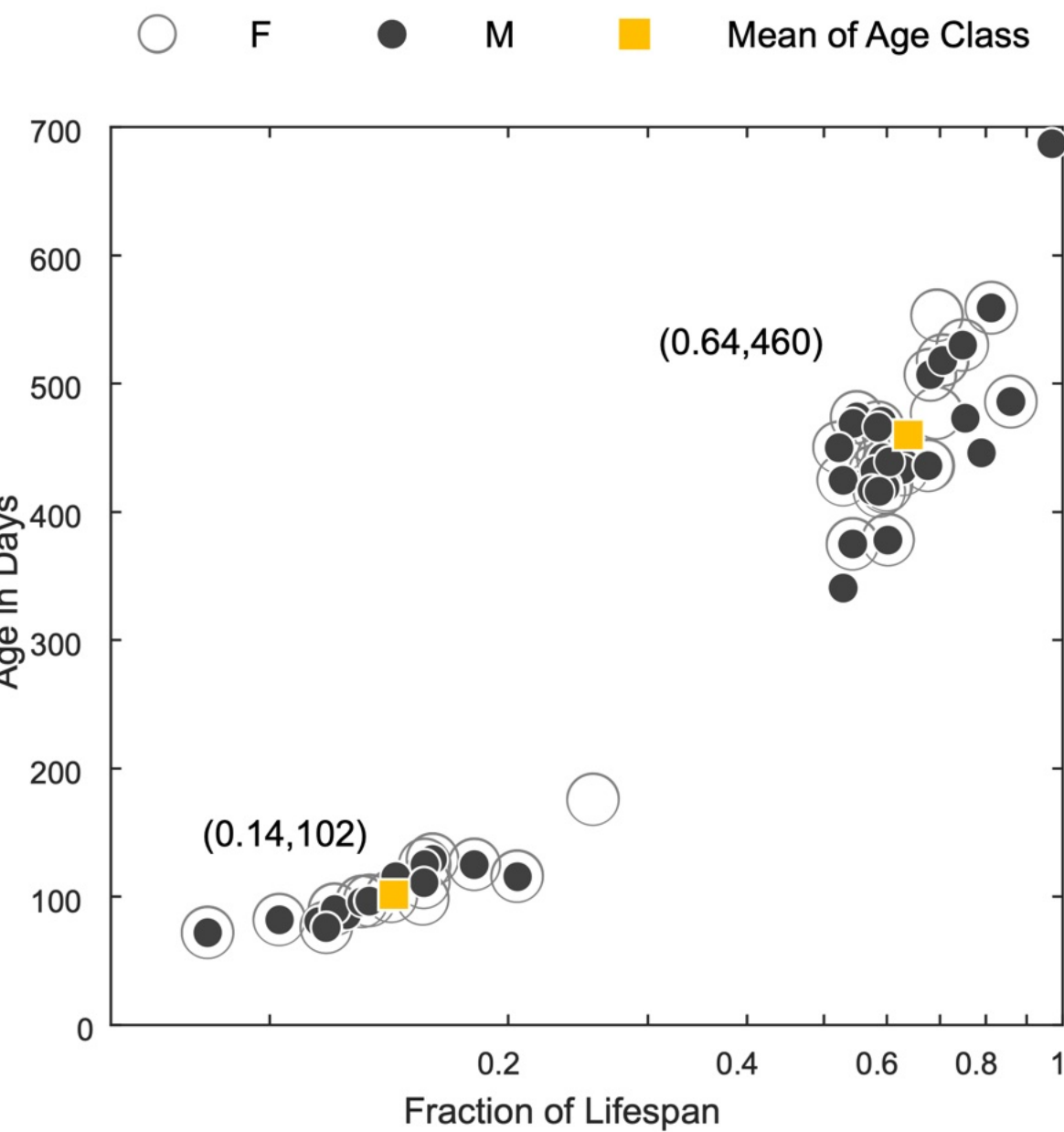


Figure 1: Estimated fraction of lifespan versus Age of Specimen in days at time of perfusion. The fraction of lifespan was calculated as each specimen age of days at perfusion divided by the mean longevity of female mice on a standard chow diet for each strain [4]. Male mice are represented by solid black dots. Female mice are represented by unfilled circles. The mean of each age class is represented by a gold square, the numbers to the left of each cohort are the mean (fraction of lifespan, Age in Days).

METHODS

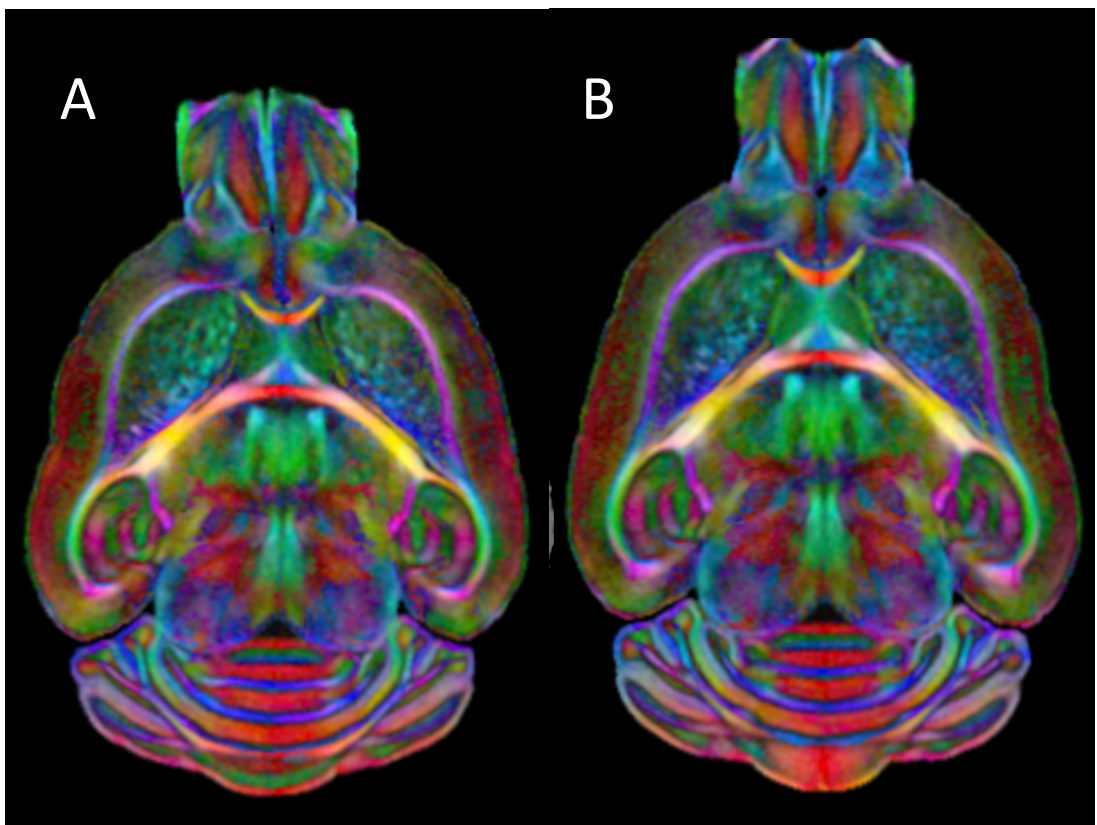
Thirteen BXD strains (Table 1) at two age classifications (Young- Mean Age At Perfusion: 14 ± 3.6% of lifespan, 102 ± 21 days, Middle-Aged- Mean Age At Perfusion: 64 ± 9.7 % of lifespan, 460 ± 59 days) were perfusion fixed with Prohance to enhance the MR signal [3]. The excised heads were scanned on the Duke 9.4T connectome scanner using a Stesjkal Tanner imaging sequence at 45 µm isotropic resolution with 46 unique gradient angles [3]. Labels were placed on 231 regions of interest (ROI) in each hemisphere [1] using an image processing pipeline to map the data to a reference atlas [5].

The mean of volume (absolute and fractional) and diffusion scalar metrics (axial diffusivity (AD), mean diffusivity (MD), radial diffusivity (RD), and fractional anisotropy (FA)) were extracted for each region. Fractional volume is where each ROI volume is normalized by the total brain volume of each mouse. Statistical processing (N-Way ANOVA for each contrast type ~ strain + sex + age class) was performed in MATLAB 2021b using custom pipelines – looking at the age class differences while controlling for sex and strain. Benjamini Hochberg (BH) FDR correction was performed considering region and contrast. We determined the estimated statistical power given Cohen's D (a directional effect size measure), the P-value, and the number of specimen in the study [6]. Connectomes were generated for each mouse in DSI Studio (version 20181116). Each connectome was normalized by a ratio of the total brain volume to a “standard” mouse with 435 mm³ brain volume. Note the size difference between mice in Figure 2. Connectomes were embedded and analyzed using Omnimanova [7] in MATLAB 2021b and R (model: N-Way MANOVA ~ strain + sex + age class, investigating age while controlling sex and strain). For global visualization, we represent the latent variables as a multi-dimensional scaling.

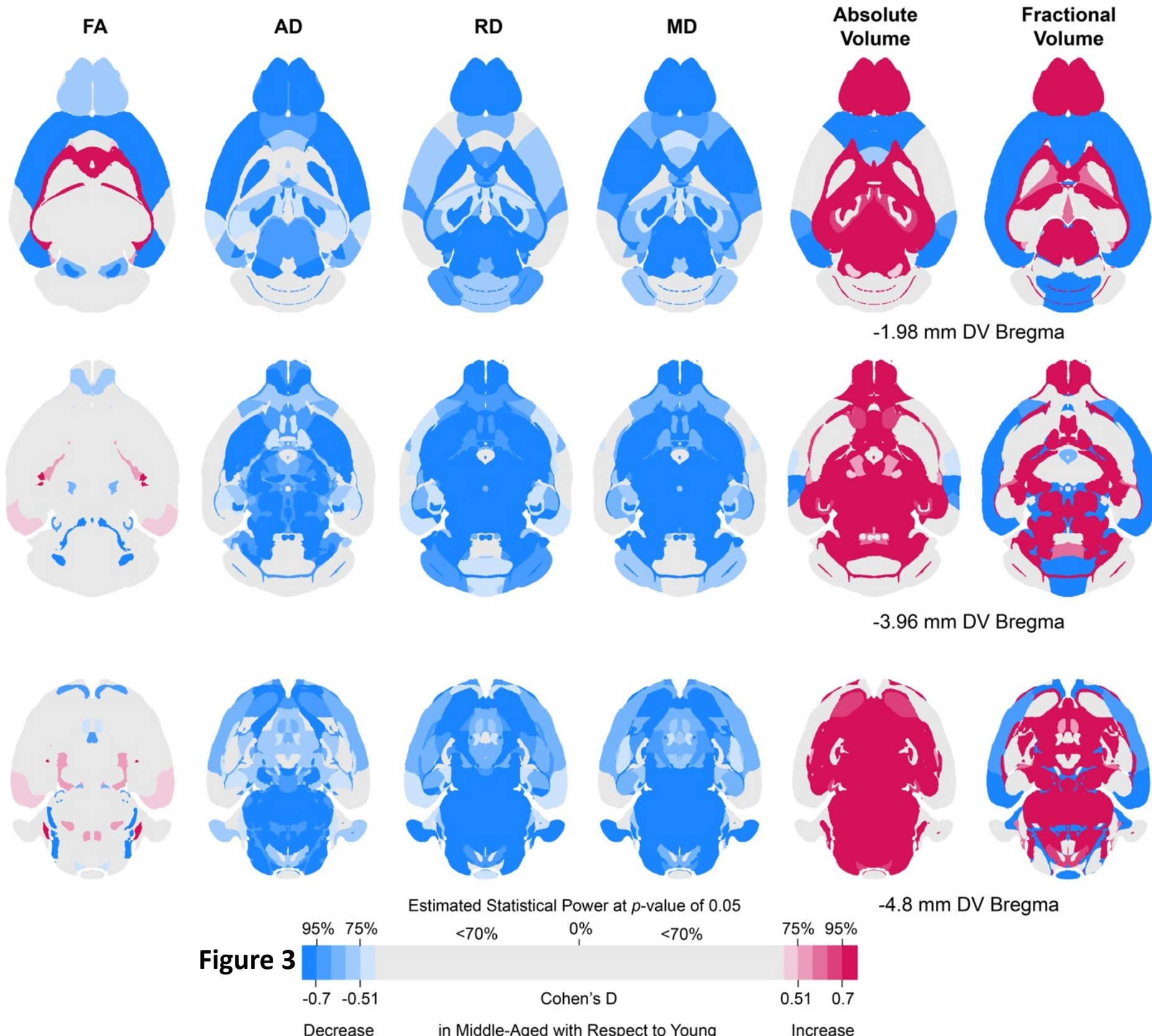
Table 1: Counts of Mice Per Strain and Age Group.

	Young	Middle Aged	Total
BXD24	4	6	10
BXD29	4	3	7
BXD34	4	4	8
BXD43	4	7	11
BXD44	4	3	7
BXD48a	4	4	8
BXD51	4	7	11
BXD60	2	4	6
BXD62	4	5	9
BXD65b	4	4	8
BXD77	4	5	9
BXD89	3	5	8
BXD101	2	3	5
Total	47	60	107

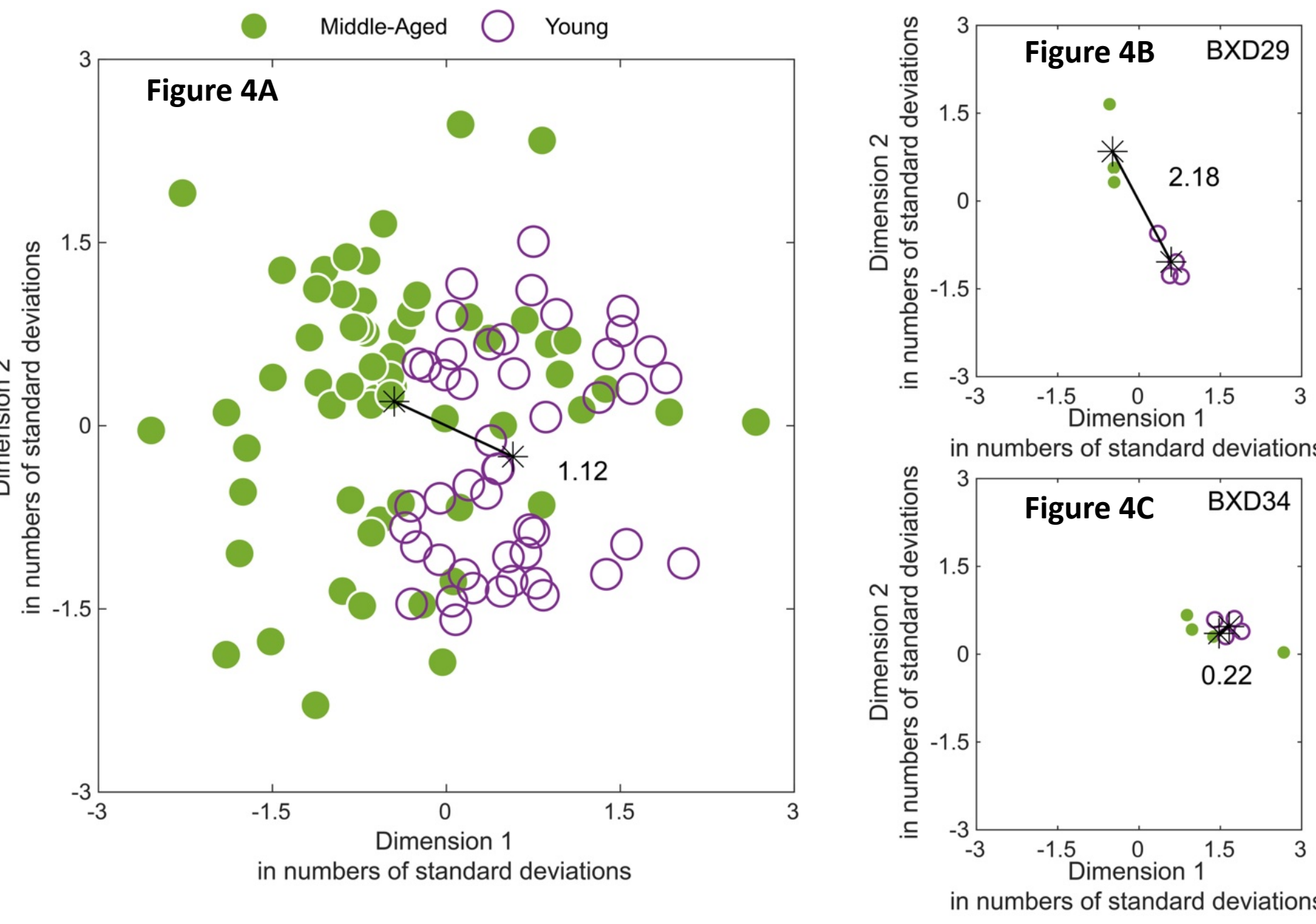
Figure 2: Representative axial color FA. From (A) 98-day female BXD60 (Young Specimen) and (B) 378-day female BXD60 (Middle-Aged Specimen).



DIFFERENTIAL AGE IMPACTS ACROSS REGIONS



GLOBAL MULTI-DIMENSIONAL SCALING OF CONNECTOME SHOWS DIFFERENTIAL STRAIN IMPACT



RESULTS

The aging BXD image phenome is a collection of scalar metric (FA, AD, RD, and MD), volume metric (absolute and fractional volume) (see Figure 3) and connectomic outputs (see Figures 4-6) we collect from dMRI. For presentation of connectome results, we reduce the number of strains to the top 3 (BXD29, BXD65b, BXD101) and the bottom 3 (BXD24, BXD34, BXD60) in relative ranking of age impact. The ranking is produced by finding the centroid-to-centroid distance of the age groups weighted by distribution, for each strain in each node. We rank strains on if they are at the top or the bottom rank in each node; counting the number of instances of a given strain at the top and bottom of rankings.

Left Column Figure Captions, Top to Bottom:

Figure 3: Three representative slices from the Duke Mouse Brain Atlas (DMBA) [1] are used to visualize global (all-strain) changes in dMRI metrics. AD, MD, and RD generally decrease with age, represented by blue shades, but volume and FA have mixed increases (red) and decreases (blue). The intensity of false coloring indicates the estimated statistical power.

Figure 4: (A) Multi-dimensional scaling plot of connectomes for all the specimens shows the global shift in connectivity with age. Subsets of all the middle-aged and young specimen in (B) (BXD29) a strain with large change and (C) (BXD34) a strain with limited change show the strain variation. Axes have been scaled such that 1 unit is equivalent to a standard deviation of the whole global dataset along that dimension. Centroid to centroid distance is marked for each graph with the distance itself included in the graph by the bar with asterisks. The shift in BXD29 is ~ 10 X that of BXD34.

EFFECT SIZE OF KEY REGIONS IN SIGNIFICANT NODES VARY ACROSS SELECT STRAINS. THE AVERAGE RESPONSE ACROSS STRAINS SHOWS A MORE MODERATED EFFECT THAN IN SELECT STRAINS.

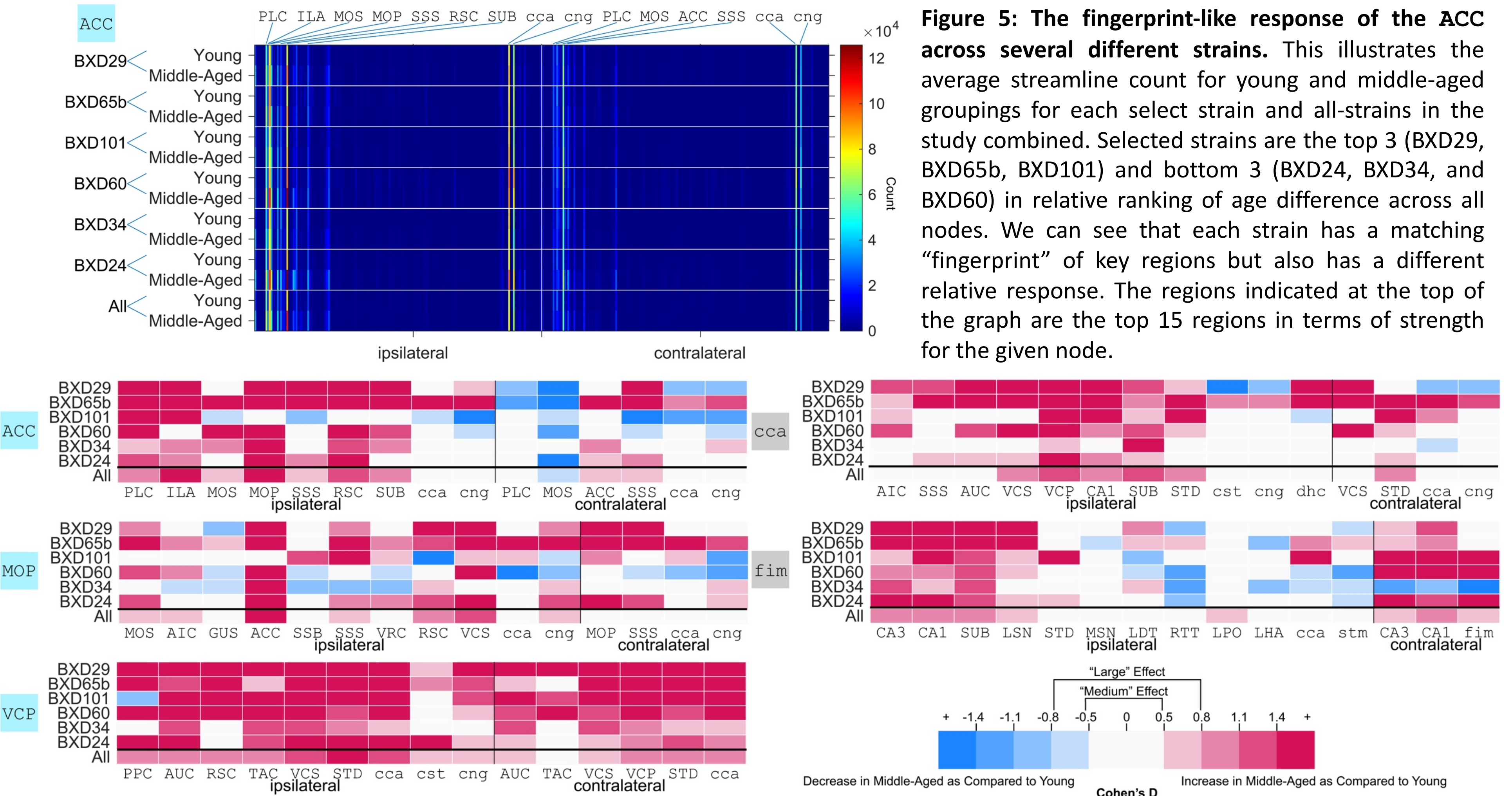


Figure 6: An expansion on Figure 5 illustrating Cohen's D Effect Size in key regions for a subset of significantly changed nodes for age difference in the cortex and white matter. This indicates that there are differences in the amount of expression for given strains and the amount of expression shifts for each node. Just because a strain is highly affected for one node does not mean it will be for another.

DISCUSSION AND CONCLUSIONS

Considering all strains, AD, RD, and MD decrease across the brain with age. Volumes generally increase in the midbrain, while decreasing in cortical areas. FA increases with age in forebrain fiber tracts while decreasing in the neocortex (outside of anterior cingulate cortex which follows the fiber tracts) and cerebellar nuclei.

The global connectomes for all the strains shift considerably (1.12) over this age range. The age effect is significant (P-value < 1e-4) accounting for 0.60 of the variance and a Cohen F² effect size of 1.22. Strain is also a major factor (P-value < 1e-4) accounting for 0.47 of the variance and a Cohen F² effect size of 0.936. The strain stratified data show the range of response with BXD29 having a 2.18 separation, while BXD34 separation of 0.22 nearly 10X smaller.

Node by node analysis of the connectome after BH correction reveals 315 nodes (88% of nodes analyzed) with corrected P-values < 0.05. We analyzed a subset of those nodes, focusing on cortex and white matter. We identified 15 edges within selected nodes as major contributors to the fingerprint for the node. Note, how the same edges appear as part of the fingerprint for ACC connectivity across all the strains. Each strain has a different degree of change, but the changed edges match across strains. An analysis of connectivity change by strain demonstrates similar trends in most of the nodes, with some interesting anomalies (ACC/BXD101, MOP/BXD60, f.i.m/BXD34).

dMRI provides a collection of image derived phenotypes that change with age. All are remarkably strain dependent, highlighting the need for use of multiple strains. These data demonstrate methods for detecting age related changes across scale from volumes, to cytoarchitecture derived from dMRI to connectomes and finally to edges laying the foundation for broad application in models of neurodegeneration.

CITATIONS AND ACKNOWLEDGEMENTS

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All MRI scanning and MRI data analysis was done within:



Center for In Vivo Microscopy

Mice were raised and behavioral data captured at:



Research Funded by: NIH R01AG070913